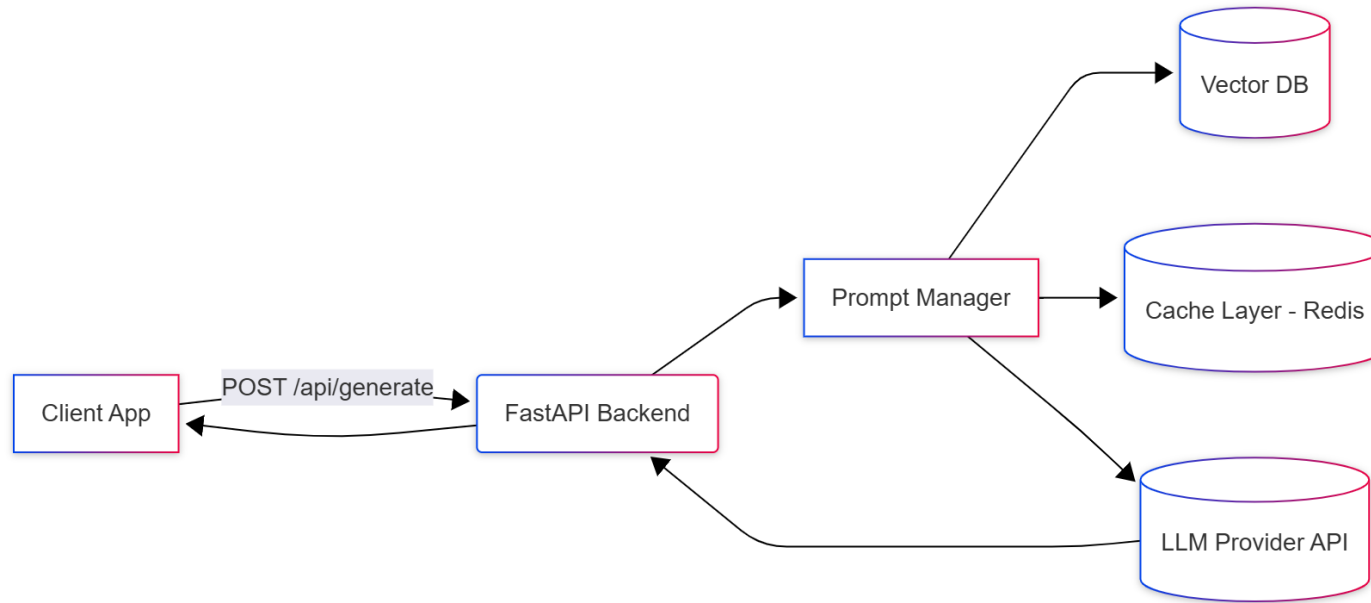
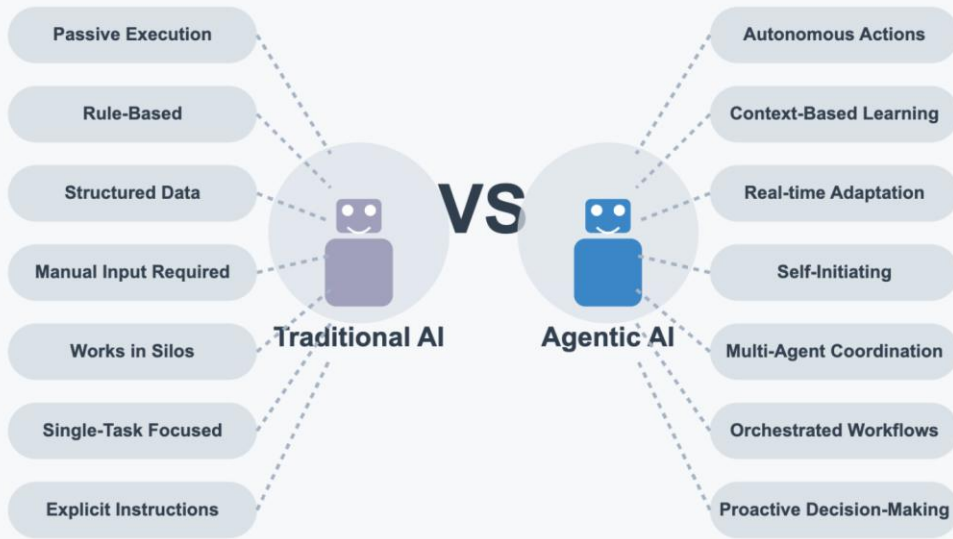




- **Goal:** Transforming AI logic into a scalable, production-ready web service.
- **Why FastAPI?** It is built on **Asynchronous Server Gateway Interface (ASGI)**, making it one of the fastest Python frameworks available, rivaling Node.js and Go.
- **Core Pillars:** Async execution, strict data validation, and real-time streaming.

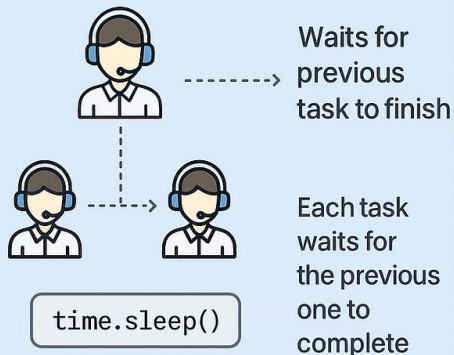
Module 3 – High-Performance AI Backend (FastAPI)

Traditional AI vs. Agentic AI

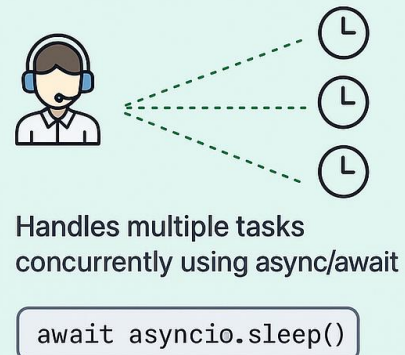


Sync vs Async Agents in Python — Blocking vs Non-blocking Execution

Sync Agent (Blocking)



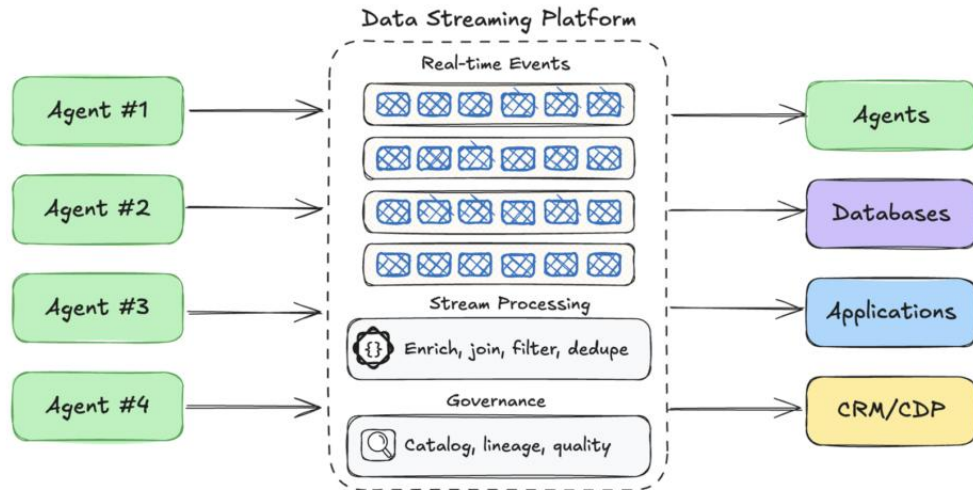
Async Agent (Non-Blocking)



Data Validation & Async Logic

- **Pydantic:** Using BaseModel to enforce strict schemas. This prevents "Type Errors" (e.g., sending a number where a string is expected) from crashing the server.
- **Asynchronous Inference:** Utilizing `async/await` to ensure the server can handle hundreds of concurrent AI requests without blocking the main thread.

Building Event-Driven Multi-Agents with Data Streaming



Real-Time Streaming (SSE)

- **Server-Sent Events (SSE):** Instead of waiting for a full 30-second AI generation, we stream tokens one-by-one.
- **User Experience:** This creates the "ChatGPT-style" typing effect, drastically improving perceived performance.
- **Auto-Docs:** FastAPI automatically generates **Swagger UI** at /docs, allowing frontend teams to test the API instantly without reading manual documentation.